DHCP-Dynamic Host Configuration Protocol

DHCP is the protocol that hands out IP addresses (IPv4/6) to devices when they first connect to a network

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Discover- Client broadcasts “I need an IP”
Offer- DHCP server replies with available IP and setting
Request- CLient requests that offered IP
Acknowlegement(ACK)- Server confirms, lease begins

DHCP Steps

1- User joins wifi. The Wifi SSID has a pre defined VLAN such as VLAN 10
2- Device sends DHCP discover broadcast. This broadcast cannot leave the VLAN and is received by the default gateway (router or Layer 3 switch).
3- The router or Layer 3 switch uses a DHCP relay (IP helper address) to forward the request to the DHCP server. The relay adds information about the originating subnet/VLAN, allowing the server to identify the correct scope.
4- The DHCP server selects the appropriate scope based on the relay information and responds with an available IP configuration.

Key DHCP Concepts (Concise)

Scope – A defined pool of IP addresses and options for a subnet
Lease – The time period an IP address is assigned to a client
DHCP Relay – Allows DHCP requests to cross VLAN boundaries
APIPA – A self-assigned address (169.254.x.x) used when DHCP fails
Common DHCP issuses include IP range exhaustion, users being given APIPA addresses, devices being placed in the wrong scope